

Uncertain Set Theory

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Abstract

This paper establishes **Uncertain Set Theory**, a novel foundational mathematical framework built from First Principles to model uncertainty without relying on probabilistic distributions, fuzzy membership degrees, or static interval bounds. Traditional approaches such as Interval Arithmetic frequently suffer from correlation loss and catastrophic numerical noise explosion, while probabilistic models require prior knowledge of probability densities that are often unavailable in real-world high-noise environments.

We address these fundamental limitations by introducing the concepts of **Uncertain Elements** and **Uncertain Sets**, defined as multi-state superposition entities characterized by possibility domains $\text{Val}(x)$ and configuration spaces $\text{Conf}(S)$. Through this scenario-based structural formulation, we demonstrate that set cardinality itself unifies naturally as an uncertain number $|S| \in \mathbf{U}(\mathbb{N})$, resolving cardinality degeneration and state-collapse seamlessly. Furthermore, algebraic operations on uncertain sets inherit classical binary set relations over their configuration spaces, preserving absolute

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logical correlation and enabling canonical complexity reduction down to $\mathcal{O}(1)$ via Lazy Evaluation. This framework serves as the core foundation for $U(\mathbb{K})$ arithmetic, scenario-space optimization, and high-dimensional uncertain topology.

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Chapter 1

Uncertain Sets

Definition 1.1 (Uncertain Element). *Let U be a non-empty universal set. An **uncertain element** x on U is an object taking an undetermined value in U , constrained by a set of possible values $A \subseteq U$ ($A \neq \emptyset$), denoted by:*

$$x \in A$$

*The set A is called the **possibility domain** (or state space) of x , and we denote $\text{Val}(x) = A$.*

Definition 1.2 (Uncertain Set). *Let U be a non-empty universal set and I be an index set. An **uncertain set** S on U is a collection of uncertain elements of the form:*

$$S = \{x_i : i \in I\}$$

where each x_i ($i \in I$) is an uncertain element on U with possibility domain $A_i = \text{Val}(x_i) \subseteq U$.

*A **realized configuration** (or empirical state) of S is a classical set $S^* \subseteq U$ obtained when the elements x_i simultaneously take specific values $v_i \in A_i$:*

$$S^* = \{v_i : i \in I\}$$

The collection of all possible configurations of S is denoted by $\text{Conf}(S)$.

Definition 1.3 (Uncertain Cardinality). *Let $S = \{x_i : i \in I\}$ be an uncertain set on the universal set U . The **uncertain cardinality** (or*

number of uncertain elements) of S , denoted by $|S|$, is an uncertain number on the set of natural numbers $\mathbb{N}$ whose possibility domain is defined by:

$$\text{Val}(|S|) = \{|S^*| : S^* \in \text{Conf}(S)\} \subseteq \mathbb{N}$$

where $|S^*|$ represents the classical cardinality of the configuration S^* .

Remark 1.1. If all uncertain elements x_i in S take distinct values across their entire possibility domains, we have $|S^*| = |I|$ for all $S^* \in \text{Conf}(S)$, in which case the uncertain cardinality degenerates into a classical number $|S| = |I|$. Conversely, size degeneration (cardinality collapse) occurs at configurations S^* where duplicate values appear among the elements ($v_i = v_j$ for $i \neq j$).

Definition 1.4 (Binary Operations on Uncertain Sets). Let S_1 and S_2 be two uncertain sets on the universal set U , and let $\star$ be a classical binary set operation (such as $\cup, \cap, \setminus$). The corresponding binary operation on uncertain sets, denoted by $S_1 \star S_2$, is an uncertain set on U whose configuration space is defined by:

$$\text{Conf}(S_1 \star S_2) = \{S_1^* \star S_2^* : S_1^* \in \text{Conf}(S_1), S_2^* \in \text{Conf}(S_2)\}$$

Example 1.1 (Uncertain Union and Intersection). From the general definition above, the union and intersection between two uncertain sets S_1 and S_2 have the following corresponding configuration spaces:

$$\begin{aligned} \text{Conf}(S_1 \cup S_2) &= \{S_1^* \cup S_2^* : S_1^* \in \text{Conf}(S_1), S_2^* \in \text{Conf}(S_2)\}, \\ \text{Conf}(S_1 \cap S_2) &= \{S_1^* \cap S_2^* : S_1^* \in \text{Conf}(S_1), S_2^* \in \text{Conf}(S_2)\}. \end{aligned}$$

Example 1.2 (Numerical Illustration and Uncertain Cardinality). Consider the universal set $U = \mathbb{R}$. Let x_1 and x_2 be two uncertain elements whose possibility domains are respectively:

$$\text{Val}(x_1) = \{1, 2\}, \quad \text{Val}(x_2) = \{1, 3\}$$

Consider the uncertain set $S = \{x_1, x_2\}$.

Assuming the states of x_1 and x_2 vary independently, the configuration space $\text{Conf}(S)$ consists of $2 \times 2 = 4$ realized states:

- When $x_1 = 1, x_2 = 1 \implies S_1^* = \{1, 1\} = \{1\}$ (due to element deduplication in classical set theory), implying $|S_1^*| = 1$.
- When $x_1 = 1, x_2 = 3 \implies S_2^* = \{1, 3\}$, implying $|S_2^*| = 2$.
- When $x_1 = 2, x_2 = 1 \implies S_3^* = \{2, 1\}$, implying $|S_3^*| = 2$.
- When $x_1 = 2, x_2 = 3 \implies S_4^* = \{2, 3\}$, implying $|S_4^*| = 2$.

Therefore, the set of all possible configurations of S is:

$$\text{Conf}(S) = \{\{1\}, \{1, 3\}, \{1, 2\}, \{2, 3\}\}$$

Hence, the uncertain cardinality $|S|$ has a possibility domain that constitutes an uncertain number on $\mathbb{N}$:

$$\text{Val}(|S|) = \{|S^*| : S^* \in \text{Conf}(S)\} = \{1, 2\} \subseteq \mathbb{N}$$